P1.

a) maximize$_{x,\Delta}$ Δ

s.t. $\sum_j x_{ij} - \sum_i x_{ji} = 0, \quad \forall\, i \neq 1, 4$

$\sum_j x_{j1} - \sum_j 1j + \Delta = 0$

$\sum_i x_{j4} - \sum_j 4j - \Delta = 0$

$x_{ij} \leq c_{ij} \qquad \forall\, ij \in E$

$x_{ij} \geqslant 0 \qquad \forall\, ij \in E$

$$C = (c_{ij})_{4\times 4} = \begin{bmatrix} 0 & 8 & 7 & 0 \\ 0 & 0 & 2 & 4 \\ 0 & 0 & 0 & 12 \\ 0 & 0 & 0 & 0 \end{bmatrix}$$

$$E = \{(1,2), (1,3), (2,3), (2,4), (3,4)\}$$

```
cvx_begin quiet
variables x12 x13 x23 x24 x34 delta;
    minimize -delta
    subject to
        x23 + x24 - x12 == 0;
        x13 + x23 - x34 == 0;
        -x12 - x13 + delta == 0;
        x24 + x34 - delta == 0;
        x12 <= 8;
        x13 <= 7;
        x23 <= 2;
        x24 <= 4;
        x34 <= 12;
        [x12, x13, x23, x24, x34] >= 0;
cvx_end

%输出
fprintf('x12: %.2f\n', x12);
fprintf('x13: %.2f\n', x13);
fprintf('x23: %.2f\n', x23);
fprintf('x24: %.2f\n', x24);
fprintf('x34: %.2f\n', x34);
fprintf('Optimal of primal: %.2f\n', -cvx_optval);

```

```
x12: 6.00
x13: 7.00
x23: 2.00
x24: 4.00
x34: 9.00
Optimal of primal: 13.00
```

optimal value is 13

b) dual problem :

$$\text{minimize} \sum_{1 \le i,j \le 4} c_{ij} z_{ij}$$

$$\text{s.t.} \quad z_{ij} \ge y_i - y_j, \quad \forall\, ij \in E$$

$$y_s - y_t = 1$$

$$z_{ij} \ge 0$$

y_is are either 0 or 1.

$x_{12} < c_{12} \quad x_{34} < c_{34}$

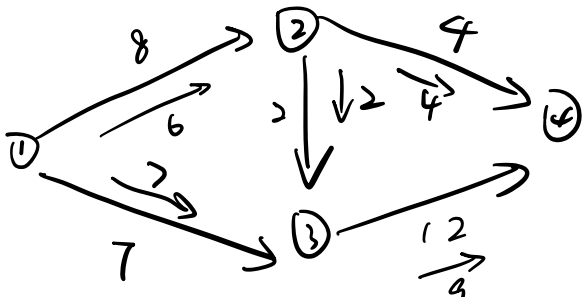
$\therefore z_{12} = z_{34} = 0$

$x_{13} - c_{13} = x_{23} - c_{23} = x_{24} - c_{24} = 0$

$\therefore z_{13} = z_{23} = z_{24} = 1$

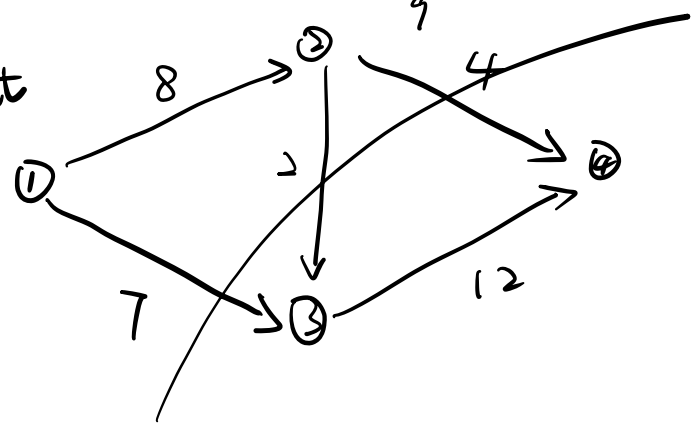
$\therefore$ optimal value is 13

c) maximum flow



then it's 13

min cut



P2

a) $x^* = (5, 15, 0, 10, 0)$

optimal value is 175

b) $A_B = \begin{bmatrix} 3 & 0 & 1 \\ 2 & 1 & 4 \\ 0 & 0 & 1 \end{bmatrix}$ $\begin{matrix} 1/3 & 0 & -1/3 \\ -2/3 & 1 & -10/3 \\ 0 & 0 & 1 \end{matrix}$

$A_B^{-1} = \begin{bmatrix} 1/3 & 0 & -1/3 \\ -2/3 & 1 & -10/3 \\ 0 & 0 & 1 \end{bmatrix}$

$\tilde{b} = \begin{bmatrix} 50+\lambda \\ 60 \\ 5 \end{bmatrix}$ $A_B^{-1}\tilde{b} = \begin{bmatrix} 15+\lambda/3 \\ 10-2\lambda/3 \\ 5 \end{bmatrix} \geq 0$

$\therefore -45 \leq \lambda \leq 15$

c) $\lambda = 10$ $A_B^{-1}\tilde{b} = \begin{bmatrix} 55/3 \\ 10/3 \\ 5 \end{bmatrix}$

$x^* = (5, \frac{55}{3}, 0, \frac{10}{3}, 0)$ $25+550$

optimal value is $\frac{625}{3}$

d) $2 \in B$ $\tilde{C}_B^T = (-10+\lambda, 0, -5)$

$\tilde{C}_N^T = C_N^T - \tilde{C}_B^T A_B^{-1} A_N$

$= (0,0) - (-10+\lambda, 0, -5) \begin{bmatrix} 1/3 & -1/3 \\ -2/3 & -10/3 \\ 0 & 1 \end{bmatrix}$

$= (\frac{10}{3} - \frac{\lambda}{3}, \frac{5}{3} + \frac{\lambda}{3}) \geq 0$

$-5 \leq \lambda \leq 10$